

1. ARP

a)

- **H1 cache:** 10.0.10.1 to MAC c Because H1 first targets router R1 because H4 is on a another /24 subnet
- **H2 cache:** 10.0.10.11 to MAC a Because ARP snooping is enabled, H2 learns this from H1's broadcasted ARP request
- **R1 cache:** 10.0.10.11 to MAC a, 10.0.20.44 to MAC f Because R1 learns H1's mapping from the request of the ARP on LAN 1 and H4's mapping from the reply of the ARP on LAN 2
- **H3 cache:** 10.0.20.1 to MAC d because ARP snooping is enabled, H3 learns this from R1's broadcasted ARP request
- **H4 cache:** 10.0.20.1 to MAC d because H4 learns this since its the target of R1's request

b)

- **B1:** Port N to MAC a, PORT E to MAC c because B1 learns MAC a when H1 does ARP request and MAC c when R1 sends its ARP reply
- **B2:** Port W to MAC d, Port S to MAC f because B2 learns MAC d when R1 sends its ARP request and MAC f when H4 sends its reply.

c)

- **H1 cache:** 10.0.10.1 to MAC c
- **H2 cache:** nothing because passive snooping is disabled so H2 drops H1's request since its not the target
- **R1 cache:** 10.0.10.11 to MAC a, 10.0.20.44 to MAC f
- **H3:** nothing
- **H4 cache:** 10.0.20.1 to MAC d

2. UDP and fragmentation

IP fragment offsets are written in 8-byte blocks. The byte size of every non-final fragment must be a multiple of 8. $\lfloor 1260/8 \rfloor \times 8 = 157 \times 8 = 1256$

a)

We simply divide the application data size by the size of the fragment and we get $\lceil 5408/1256 \rceil = 5$. So 5 **fragments** are transmitted

b)

Fragment ID	Payload	Total Length	MF bit	Fragment offset
Frag 1	Bytes 0-1255	1276 bytes	1	0 (0/8)
Frag 2	Bytes 1256-2511	1276 bytes	1	157 (1256/8)
Frag 3	Bytes 2512-3767	1276 bytes	1	314 (2512/8)

Fragment ID	Payload	Total Length	MF bit	Fragment offset
Frag 4	Bytes 3768-5023	1276 bytes	1	471 (3768/8)
Frag 5	Bytes 5024-5407	404 bytes	0	628 (5024/8)

3. Routing

a)

Destination	Metric	Next hop(s)
192.168.1.0/24	2	192.168.2.1, 192.168.3.1
192.168.2.0/24	1	-
192.168.3.0/24	1	-
192.168.4.0/24	1	-
192.168.5.0/24	2	192.168.3.1

b)

- (i)
 - For the packet to 192.168.1.88 (net1) Router C is allowed to use 192.168.2.1 and 192.168.3.1
 - For the packet to 192.168.5.140 (net5) Router C is only allowed to use 192.168.3.1
 - Why?: Router C supports equal-cost multi-path (ECMP) routing. It splits traffic for network 1 because it has two equally good paths with a metric of 2, but it cannot split traffic at net5 because the path via Router A (metric 2) is better than the alternative path via Router B (metric 3)
- (ii) It gives the network load balancing which improves network throughput and reduces congestion

5. TCP

Parameters of the network:

transmission_time: (1200 Bytes / 960,000 bps) = 10ms

propagation_time: one way propagation time is 50ms

segment_arrival_time: sent_time + transmission_time + propagation_time
 = sent_time + 10ms + 50ms = sent_time + 60ms

MSS: Path MTU (1240 bytes), IPV4 Header (20 bytes), TCP Header (20 bytes)

Total segments: (13200 bytes / 1200 bytes per segment) = 11
segments total

a)

1. RTT

- host A sends **seg 1** at $t = 0$. Transmission ends at $t = 10$
- **seg 1** arrives at Host B at $t = 60$ ms
- Host B delays ACK for 300ms because only one was recieved
- ACK 2200 arrives at A at $360 + 50 = 410$ ms so from the start of the transmission the first RTT is 410 ms

2. RTO Calculation (RFC 6298):

- SRTT = 410 ms.
- RTTVAR = $R / 2 = 205$ ms.
- The clock granularity G is 250 ms (0.25 seconds).
- $RTO = SRTT + \max(G, 4 * RTTVAR) = 410 + \max(250, 4 * 205) = 410 + 820 = 1230$ ms.

3. $t_1 = 410$ ms + 1230 ms = 1640 ms.